Aarush Ram Anandh

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Hiring Team
Sony Research India

Application: Speech Recognition Internship

Dear Hiring Team,

I am applying for the Speech Recognition internship at Sony Research India. I am completing a B.Sc. (Hons.) in Computer Science and Mathematics at Ashoka University, expected in 2028. My most relevant project is MEIKURAL, an ongoing Hindustani raag-recognition system that has given me practical experience with audio preprocessing, spectrogram-based learning and inference diagnostics.

MEIKURAL classifies seven raags using the Saraga Hindustani dataset. I implemented resampling, RMS normalisation, overlapping 30-second windows and STFT features, and explored per-window tonic alignment, full-recording alignment and native-frequency representations. I trained a native-STFT CNN on 3,471 windows over 30 CUDA epochs and built interactive clip inference, prediction timelines and recording-level and window-level confusion-matrix reports. This work has made representation choices, input handling and error inspection central parts of how I approach audio models.

My wider research includes a public preprint on inference-time colour preservation in latent diffusion and ongoing controllable optimal-transport research. At the Koita Centre for Digital Health, I contributed to a wearable-data pipeline achieving approximately 70% sensitivity for asthma exacerbation detection in study evaluation. My work processes 12.1 million HR and 736,000 HRV readings into smoothing, residual and change-point features, with patient-wise evaluation. I have also built a webcam-tracking prototype that streams coordinates and segmentation masks through Flask-SocketIO to a browser interface, giving me experience connecting an algorithm to an interactive feedback loop.

At Avartan Labs, I raised rare-class correct detection from 40% to 80% in an RT-DETR pipeline covering 18 material classes. This industry internship gave me experience analysing detector failures and investigating targeted model improvements.

I would like to build on this foundation in speech recognition and speaker diarization, contributing Python/PyTorch implementation, signal-processing work and careful evaluation. The chance to learn from Sony's speech researchers while working toward useful real-time systems is especially attractive. Thank you for considering my application.

Yours sincerely,

Aarush Ram Anandh